

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

CHEMISTRY

Paper 2 Structured Questions

9647/02

TUESDAY 13 SEPTEMBER 2011

2 Hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

A Data Booklet is provided

You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part of the question.

At the end of the examination, fasten all work securely together.

For Examiner's Use		
Paper 1		
Paper 2	Q 1	12
	Q 2	18
	Q 3	18
	Q 4	12
	Q 5	12
		72
Paper 3		
Total	192	

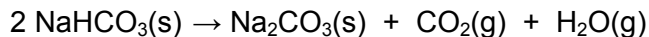
This document consists of **17** printed pages and **1** blank page.

[Turn over]

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1 Planning (P)

A mixture of sodium carbonate, Na_2CO_3 and sodium hydrogencarbonate, NaHCO_3 is heated strongly with a bunsen burner. **Only** sodium hydrogencarbonate, NaHCO_3 decomposes on heating.



You are to design an experiment to determine the percentage by mass of sodium hydrogencarbonate, NaHCO_3 in a 5.00 g sample of a mixture of sodium carbonate, Na_2CO_3 and sodium hydrogencarbonate, NaHCO_3 , by **heating** alone.

The only apparatus available consists of a boiling-tube and holder, a heat-proof mat, a chemical balance and a bunsen burner.

(a) Outline step by step, the practical sequence for the method you would use to

- make appropriate weighings,
- decompose NaHCO_3 in the sample by heating,
- ensure that decomposition was complete.

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- (b)** Show how you would tabulate your results. Include in your table and other masses you would calculate from the experimental results to enable you to determine the percentage by mass of NaHCO_3 in the sample of the mixture. Insert in your table, the letters **A**, **B**, **C** etc. to represent each measurement of mass.

[3]

(c) Use the letters you have entered in (b) to show how you would process the results to find

(i) the mass of NaHCO_3 in the sample of the mixture

(ii) the % by mass of NaHCO_3 in the sample of the mixture.

[4]

(d) Suggest an alternative experimental method, **not** involving weighing, which could be used to determine the % by mass of NaHCO_3 in the sample of the mixture.

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..... [1]

[Total: 12]

- 2 Most automobiles today are powered by internal combustion engine, where a certain fuel is used and burned with a mixture of air to produce useful energy. In Singapore, methane (marketed as compressed natural gas, CNG), has been actively promoted and used as an alternative fuel to petrol (contains mainly iso-octane). Biofuels such as methanol and ethanol have been considered viable alternative fuels as well. Some properties of the fuels are listed in the table below.

Types of Fuel	$\Delta H_c^\ominus$ Standard enthalpy change of combustion (kJ mol ⁻¹)	Boiling point (°C)	Solubility in water (g/cm ³)
CH ₄	-891	-161	35
iso-octane (C ₈ H ₁₈)	-5461	99	insoluble
CH ₃ OH	-726	60	very soluble
CH ₃ CH ₂ OH	-1300	74	very soluble

- (a) Iso-octane has the IUPAC name 2,2,4-trimethylpentane. Draw the structural formula of iso-octane. (You may wish to use skeletal formula.)

[1]

- (b) Boiling point is an important practical consideration when selecting fuel.

- (i) Account for the difference in boiling point between iso-octane and methane.

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- (ii) Suggest a possible benefit of using petrol (iso-octane) compared to methane as fuel due to its boiling point.

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[3]

(c) Another important factor for suitable alternative fuel is the amount of heat released which can be related to the standard enthalpy change of combustion, ΔH_c of the compound.

(i) Define the term standard enthalpy change of combustion.

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(ii) From the data given in the table, ΔH_c of methanol is -726 kJ mol^{-1} . However, when the value is calculated from bond energy data, the ΔH_c of methanol is -886 kJ mol^{-1} . Account for the large difference in the values.

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(iii) With reference to the table, iso-octane releases the most energy. But for practical reasons, the most ideal fuel is best quantified by the highest energy released per unit mass. By considering this, state which fuel is most ideal and energy released per gram.

Fuel:

Energy evolved: kJ g^{-1}

(iv) A common fear of installing CNG tank is the possible leakage and explosion of the methane gas. However, methane will only burn at 1200 K without any spark or ignition. Suggest the reason for the need of a high temperature.

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[5]

(d) The use of combustion to generate energy results in undesirable and harmful gas released in the exhaust. Due to the nature of the fuel and insufficient oxygen in the air, incomplete combustion may occur. Furthermore, high temperature of the engine may cause reaction between normally unreactive gases like oxygen and nitrogen.

(i) State the gas released due to incomplete combustion of carbon-based fuel such as methane and its harmful effect at high concentration.

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(ii) Alcohols have an advantage over hydrocarbons, as they have lower chances of incomplete combustion. Suggest why this is so from their molecular formulae.

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(iii) Production of nitrogen oxides can lead to an increase in greenhouse effect as well. Besides that, suggest another environmental impact of these oxides.

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[3]

(e) Water in the fuel is undesirable as its presence reduces the heat energy obtained from combustion as some energy is used to evaporate the water. Hence, identify which fuel is a poor choice due to this. Briefly explain your answer in terms of bonding.

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..... [2]

(f) The use of the fuel cells is an alternative method to power an automobile. The same chemical reaction occurs as that of the combustion between methane and oxygen, producing carbon dioxide and water. The useful energy generated for a fuel cell is the Gibbs free energy instead.

(i) Calculate the standard Gibbs free energy change, $\Delta G^\ominus$ of methane, given that standard entropy change of combustion is $-245 \text{ J K}^{-1} \text{ mol}^{-1}$.

(ii) State the oxidation numbers of carbon in methane and carbon dioxide, and the number of electrons transferred in the reaction.

Methane: Carbon dioxide: Number of electrons:

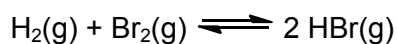
(iii) $\Delta G^\ominus \times 1000 = -nFE^\ominus$ is an equation that can relate $\Delta G^\ominus$ (in kJ mol^{-1}), and $E^\ominus$ (in V). By considering the answer in **(ii)** and given that n is the number of moles of electrons transferred while F is the Faraday constant, calculate the $E^\ominus$ for a methane/oxygen fuel cell.

[4]

[Total: 18]

- 3** Hydrobromic acid, HBr, is one of the strongest mineral acids known and is even stronger than hydrochloric acid. Both the industrial and laboratory syntheses of HBr are well documented due to the usefulness of HBr in many chemical reactions.

(a) The primary industrial preparation of HBr involves the platinum-catalysed reaction between hydrogen and bromine at temperatures between 450 – 700 K.



The reaction was carried out in a 250 m³ vessel at 607 K.

- (i)** Calculate the amount of HBr present at equilibrium, given that the equilibrium partial pressure of HBr is 2 atm.

At 607 K, the equilibrium constant, K_p , for the above reaction is 2.50.

- (ii)** Write the expression for the equilibrium constant, K_p , for the above reaction. #

- (iii)** At equilibrium, the number of hydrogen molecules is ten times that of bromine molecules.

Calculate the amount of bromine present at equilibrium.

- (iv) Assuming that only hydrogen and bromine were present at the start of the reaction, hence, determine the mass of bromine (in kg) required.

[7]

- (b) The preparation of the acid, HBr in the laboratory can be carried out by the reaction between bromine, sulfur dioxide and water **only**. The only by-product formed is also a strong acid.

- (i) Write the equation for the above laboratory preparation of HBr.

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- (ii) When concentrated solutions of the two products formed in the above preparation reacts with each other, Br_2 is regenerated. State and explain the type of reaction that is occurring.

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[3]

(c) Another method of producing HBr in the laboratory is carried out by reacting sodium bromide with concentrated phosphoric(V) acid, H_3PO_4 .

(i) Give the equation for the reaction between sodium bromide and concentrated phosphoric(V) acid.

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(ii) Suggest a reason why this method is preferred in the laboratory over the method described in part (b).

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[2]

(d) Intra and intermolecular bondings can explain the trends in the thermal stability and boiling points of the hydrogen halides.

(i) State and explain the trend in the boiling points of hydrogen halides from HF to HI.

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(ii) State and explain the trend in the thermal stability of hydrogen halides from HCl to HI.

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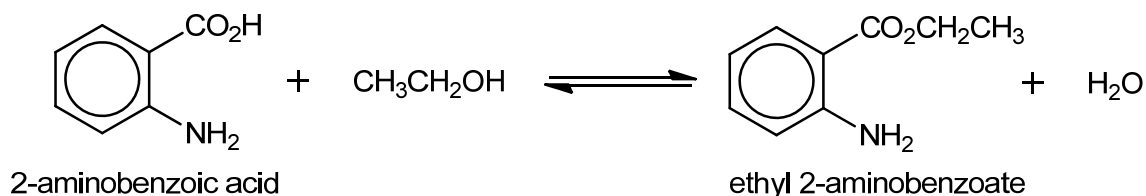
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[6]

[Total: 18]

- 4 Ethyl 2-aminobenzoate is an ester that is commonly used as food flavouring to give it a grape smell. The ester can be synthesized by adding 2-aminobenzoic acid and ethanol in a round-bottom flask and heating it under reflux. At the end of the reaction, two immiscible layers are obtained.



[1]

..... [1]

By-product:

Equation:

(d) After the reaction, aqueous sodium hydroxide, which is a **strong** base, was added to the mixture of two immiscible layers to remove some impurities. The aqueous layer was discarded while the organic layer was heated and then distilled.

(i) Suggest the two species that are removed in the aqueous layer.

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(ii) Explain with the use of an equation why a small amount of aqueous sodium hydroxide left in the organic layer would result in a lower yield of ester and state the type of reaction that is occurring.

[4]

(e) Another ester, which is an isomer of ethyl 2-aminobenzoate, **cannot** be synthesized using the method described above.

(i) State the type of isomerism and draw the structure of the isomer.

Type of isomerism:

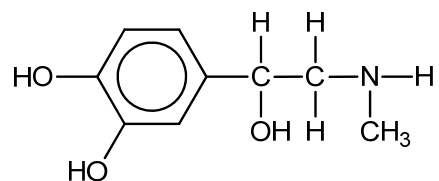
Structure of isomer:

(ii) Hence, outline the synthesis for the ester drawn in (e)(i) starting from a suitable carboxylic acid.

[4]

[Total: 12]

- 5 **Adrenaline** is a natural hormone and acts as a stimulant when secreted directly into the bloodstream. It has the following structure:



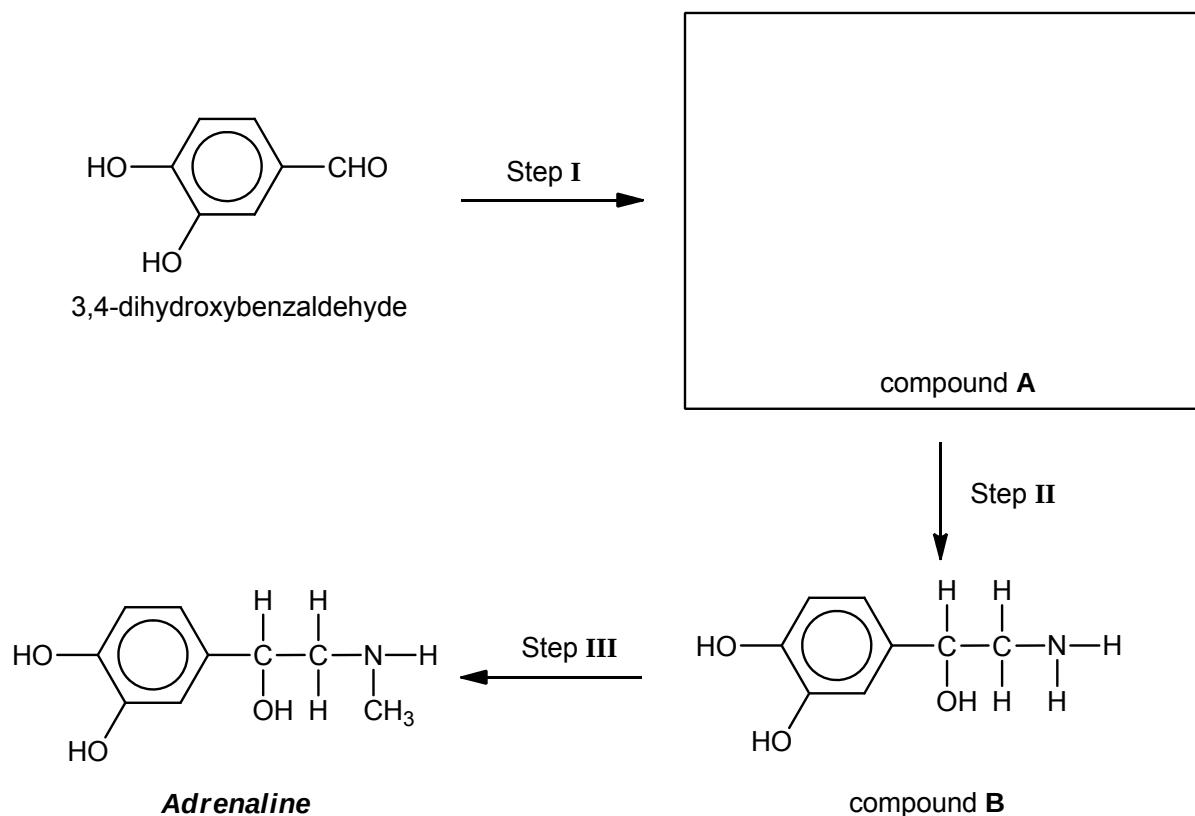
- (a) In the boxes below, give the structural formulae of the organic compounds formed when **Adrenaline** reacts with each of the following reagents and under the stated conditions.

Reagents and conditions	Product
(i) SOCl_2	
(ii) CH_3COOH , r.t.p	
(iii) Na	

[4]

- (b) Reactions involving the formation of a hydroxynitrile (cyanohydrin) as an intermediate are useful because of the further reactions that can be carried out. More importantly, it increases the number of carbon atoms in a chain as a C-C bond is formed.

Adrenaline can be synthesised from 3,4-dihydroxybenzaldehyde via the following reaction pathway with a hydroxynitrile being the intermediate.



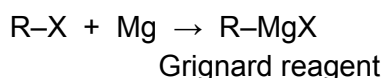
- (i) Draw the **displayed formula** of compound **A** in the box provided above.
- (ii) State the reagents and conditions required for Steps **I – III** by completing the table below.

Steps	Reagents and conditions
I	
II	
III	

[4]

- (c) The Grignard reaction is another versatile tool for forming C-C bonds. In 1900, Victor Grignard reported that a warm solution of bromoethane can react with magnesium in a dry, non-polar solvent to give an organomagnesium compound with formula of $\text{CH}_3\text{CH}_2\text{MgBr}$. The carbon-magnesium bond is polarised and the carbon directly bonded to magnesium is **basic**.

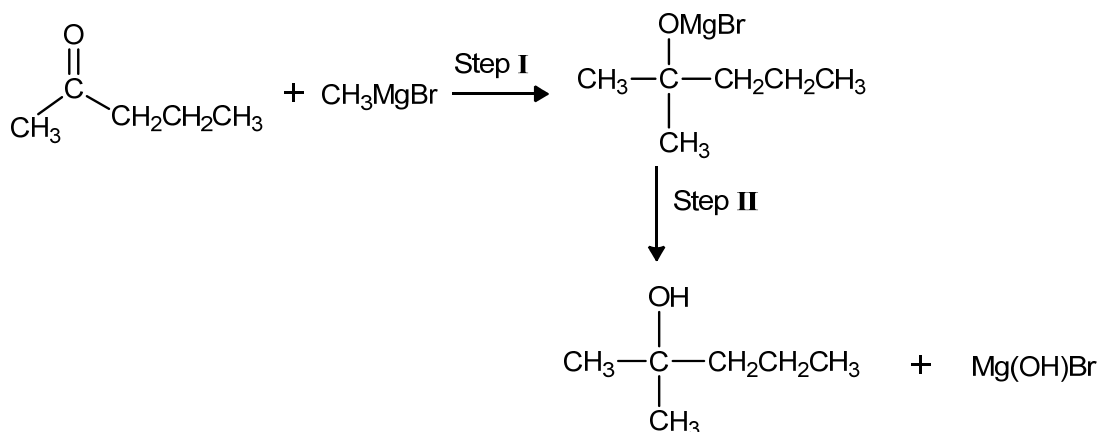
Grignard reagents, with different alkyl or aryl groups can be prepared and are widely used in organic synthesis.



(where R is an alkyl or aryl group and X can be Cl, Br or I)

The Grignard reagent formed can then react with aldehydes and ketones to yield secondary and tertiary alcohols respectively.

An example of the use of Grignard reagent is the two-step reaction of pentan-2-one with CH_3MgBr .



- (i) Identify the types of reactions that occur in Steps I and II.

Step I:

Step II:

- (ii) Draw the structure of the final organic compound formed when benzaldehyde reacts with $\text{CH}_3\text{CH}_2\text{MgBr}$.

- (iii) Suggest why this method of using the Grignard reagent is **not** suitable for the preparation of **Adrenaline** from 3,4-dihydroxybenzaldehyde.

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[4]

[Total: 12]